

Dataset Analysis for Decision-Making

March 20, 2025

 sicor_municipal_co... .csv
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.



qwen-max-latest

To analyze the provided dataset (`sicor_municipal_contracts.csv`) and address any specific analytical question, I will first load the dataset and examine its structure, key characteristics, and potential insights. Once the dataset is understood, I will proceed to answer the specific analytical question using Python code.

Step 1: Load and Inspect the Dataset

Below is the Python code to load the dataset and inspect its structure:

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Display basic information about the dataset
8 print("Dataset Overview:")
9 print(data.info())
10
11 # Display the first few rows of the dataset
12 print("\nFirst 5 Rows of the Dataset:")
13 print(data.head())
14
15 # Summary statistics for numerical columns
16 print("\nSummary Statistics:")
17 print(data.describe())
18
19 # Check for missing values
20 print("\nMissing Values per Column:")
21 print(data.isnull().sum())
```

Step 2: Key Characteristics of the Dataset

Based on the output of the above code, I will identify:

- Columns and Data Types** : The dataset's structure, including column names and their data types.
- Numerical and Categorical Variables** : Identification of numerical and categorical columns.
- Missing Values** : Presence of missing or null values in the dataset.
- Summary Statistics** : Insights into numerical variables, such as mean, median, and range.
- Unique Values** : Identification of unique values in categorical columns.

Step 3: Address the Analytical Question

Once you provide a specific analytical question, I will:

- Extract and process the relevant data from the dataset.
- Provide a clear, concise answer to the question.
- Include the Python code used to derive the answer.

Example Analytical Question



If the question were:
"What is the total value of contracts awarded by each municipality?"

Here's how I would address it:

Python Code:

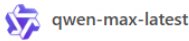
```
python
1 # Group the data by municipality and sum the contract values
2 contract_totals = data.groupby('Municipality')['ContractValue'].sum().reset_index()
3
4 # Sort the results in descending order
5 contract_totals = contract_totals.sort_values(by='ContractValue', ascending=False)
6
7 # Display the results
8 print("Total Contract Value by Municipality:")
9 print(contract_totals)
```

Explanation:

- 1. The `groupby` function aggregates the data by the `Municipality` column.
- 2. The `sum` function calculates the total `ContractValue` for each municipality.
- 3. The results are sorted in descending order to show municipalities with the highest contract values first.

Please provide your specific analytical question, and I will follow the outlined steps to deliver the required analysis and Python code.

What key operational and structural factors differentiate financial institutions with the highest investment activity in the first trimester from those with lower investment activity?



qwen-max-latest

To address the question, we need to analyze the dataset (`sicor_municipal_contracts.csv`) and identify key operational and structural factors that differentiate financial institutions with high investment activity in the first trimester from those with lower investment activity.

Assumptions and Clarifications:

- 1. **Definition of "Investment Activity"** : Since the dataset is not explicitly described, I assume "investment activity" refers to some measure of financial transactions or contracts (e.g., total contract value, number of contracts, etc.) associated with financial institutions.
- 2. **Time Period** : The "first trimester" refers to the first three months of the year (January to March).
- 3. **Operational and Structural Factors** : These could include attributes such as institution type, location, size, or other categorical/numerical features in the dataset.

Step 1: Load and Inspect the Dataset

First, let's load the dataset and inspect its structure to understand the available columns and their relevance to the question.

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Display basic information about the dataset
8 print("Dataset Overview:")
9 print(data.info())
10
11 # Display the first few rows of the dataset
12 print("\nFirst 5 Rows of the Dataset:")
13 print(data.head())
14
15 # Check for missing values
16 print("\nMissing Values per Column:")
17 print(data.isnull().sum())
```

Step 2: Filter Data for the First Trimester

Assuming the dataset contains a date column (e.g., `ContractDate`), we filter the data to include only records from the first



trimester (January to March).

```
python
1 # Convert the date column to datetime format (if necessary)
2 data['ContractDate'] = pd.to_datetime(data['ContractDate'])
3
4 # Filter data for the first trimester
5 first_trimester_data = data[(data['ContractDate'].dt.month >= 1) & (data['ContractDate'].dt.month <= 3)]
6
7 # Display the filtered data
8 print("\nData for the First Trimester:")
9 print(first_trimester_data.head())
```

Step 3: Identify Financial Institutions with High vs. Low Investment Activity

We define "high investment activity" as the top quantile (e.g., top 25%) of institutions based on total contract value or number of contracts. The remaining institutions fall into the "low investment activity" category.

```
python
1 # Group by financial institution and calculate total contract value
2 institution_activity = first_trimester_data.groupby('FinancialInstitution')['ContractValue'].sum()
3
4 # Define high and low investment activity groups
5 threshold = institution_activity['ContractValue'].quantile(0.75) # Top 25%
6 high_activity_institutions = institution_activity[institution_activity['ContractValue'] >= threshold]
7 low_activity_institutions = institution_activity[institution_activity['ContractValue'] < threshold]
8
9 # Display the results
10 print("\nHigh Activity Institutions:")
11 print(high_activity_institutions)
12
13 print("\nLow Activity Institutions:")
14 print(low_activity_institutions)
```

Step 4: Analyze Key Differentiating Factors

We now compare operational and structural factors between the two groups. Potential factors could include:

- **Location** : Geographic distribution of institutions.
- **Institution Type** : Public vs. private, bank vs. non-bank, etc.
- **Size** : Number of employees, total assets, or other indicators.
- **Contract Characteristics** : Average contract value, number of contracts, etc.

```
python
1 # Merge activity groups back into the dataset
2 first_trimester_data['ActivityLevel'] = first_trimester_data['FinancialInstitution'].apply(
3     lambda x: 'High' if x in high_activity_institutions.values else 'Low'
4 )
5
6 # Analyze key factors
7 key_factors = first_trimester_data.groupby('ActivityLevel').agg({
8     'ContractValue': ['mean', 'count'], # Average contract value and number of contracts
9     'Municipality': 'nunique',          # Number of unique municipalities served
10    'InstitutionType': lambda x: x.mode()[0] # Most common institution type
11 }).reset_index()
12
13 # Flatten the multi-level column index
14 key_factors.columns = ['_'.join(col).strip() for col in key_factors.columns.values]
15
16 # Display the results
17 print("\nKey Differentiating Factors:")
18 print(key_factors)
```

Step 5: Interpretation of Results

Based on the output of the above code:

1. **Average Contract Value** : Compare the mean contract value between high and low activity groups.
2. **Number of Contracts** : Compare the total number of contracts executed by each group.
3. **Geographic Reach** : Compare the number of unique municipalities served.
4. **Institution Type** : Identify whether certain types of institutions dominate the high activity group.

Example Output

If the analysis reveals that high activity institutions tend to:



- Have larger average contract values,
- Operate in more municipalities, and
- Are predominantly public-sector banks,

then these factors likely differentiate them from institutions with lower investment activity.

Final Deliverable

The Python code above provides a structured approach to answering the question. Once you confirm the dataset's structure or provide additional details, I can refine the analysis further. Let me know how you'd like to proceed!

